

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2024
Programme: BE Full Marks: 100
Course: System Administration in IT Infrastructure Services Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define System Administration. Discuss the importance of System Administration in IT Infrastructure. 7
b) What do you mean by Service Level Agreements and its types. Explain IT service management Frameworks. 8
2. a) Discuss the structure of the Linux file system. Explain the concepts of user and group management in Linux. 8
b) Provide an overview of Windows Server Operating Systems and explain the role of Active Directory and Domain Controllers. 7
3. a) What is subnetting, and why is it important in network management? Given the IP address 192.168.100.0 with a subnet mask of 255.255.255.0 (/24), create 4 subnets. Calculate the range of each subnet and determine the Network address and Broadcast address for each subnet. 7
b) Describe the four layers of the TCP/IP model and their respective functions. How does the TCP/IP model differ from the OSI model, and what similarities do they share? Explain in brief. 8
OR
Explain the principles of information security and identify common security threats and vulnerabilities.
4. a) List out the differences between virtualization and physical infrastructure. Explain in brief about different types of virtualizations. 7
b) Explain the different Cloud computing models and services in brief. 8
5. a) Describe the challenges of virtualization in modern IT environments. How does virtualization contribute to cost savings and improved resource allocation? Explain in brief. 8
b) Define DevOps and explain its core principles and practices. 7

6. a) How can you automate routine tasks using any of the scripting language? Explain with an example. 8
b) How can organizations design and apply a structured troubleshooting workflow to improve system administration practices? Explain. 7
7. Write short notes on: (Any two) 2×5
a) IT Service Management with ITIL
b) Domain Name System (DNS)
c) Ubuntu

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Attempt all the questions.

1. a) What are the key roles and responsibilities of a system administrator? Explain in detail. 7
b) Explain how the Service Level Agreements (SLAs) and IT Service Management (ITSM) frameworks contribute to delivering high-quality IT services. 8
2. a) Explain the different types of the Linux file system. 8

OR

Illustrate the concepts of user and group management in Linux and the importance of file permissions and security.

- b) Explain the role of Active Directory and Domain Controllers. Also, discuss the various Windows Server roles and features. 7
- 3. a) Define Protocol. Discuss the function of OSI and compare with TCP/IP protocol stack. 7

OR

Explain the principles of information security and highlight common security threats and vulnerabilities.

- b) Describe the best practices for protecting services and networks in detail. 8
- 4. a) Explain the fundamental concepts and functions of file server in detail. 7
b) Compare virtualization with physical infrastructure. Explain any two popular virtualization platforms. 8
- 5. a) What is cloud computing? Discuss the different SPI models with appropriate examples. 8
b) Explain the role of scripting languages such as Bash and PowerShell in automating routine tasks. 7

6. a) Explain how DevSecOps principles and practices enhance the secured software development and deployment lifecycle. 7
b) How can organizations design and apply a structured troubleshooting workflow to improve system administration practices? Explain in detail. 8
7. Write short notes on: (Any two) 2×5
a) Qualities of Enterprise System Admins
b) Linux Distribution
c) Type 1 and 2 Hypervisors

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Attempt all the questions.

1. a) What is System Administration? Explain the roles and responsibilities of a system administrator, providing a relevant example to support your explanation. 8
- b) What is SLA? Explain why SLA is required to meet IT services. 7
2. a) Explain Linux File System Structure. 8
- OR**
- How do we perform user and group management in linux.
- b) What is active directory and domain controller? Discuss windows server roles and features. 7
3. a) Explain OSI Reference Model and compare with TCP/ IP model. 7
- OR**
- Illustrate various security risks and vulnerabilities found in IT systems.
- b) Define Information Security. Explain its core principles and describe best practices for securing servers and networks, supported by a relevant example. 8
4. a) Describe the various types of network topologies and models. Discuss the core concepts and functions of a DNS server, providing a suitable example to support your explanation. 7
- b) Define virtualization. Compare virtualization and cloud computing. 8
5. a) Explain the various cloud service models, providing suitable examples for each. 8
- b) Discuss the function of scripting languages like Bash and PowerShell in streamlining repetitive tasks through automation. 7
6. a) Explain how Devops principles and practices help in the software development and deployment life cycle. 8

- b) What is troubleshooting? Explain a troubleshooting scenario for a server down. 7

7. Write short notes on: (Any two) 2×5

- a) Types of IT services
- b) Ubuntu
- c) VMware

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Attempt all the questions.

1. a) Define *system administration*. Discuss the primary roles and responsibilities of a system administrator, and provide a relevant example to support your discussion. 8
- b) Identify and explain the key elements of IT infrastructure services. Critically discuss the main components of Service Level Agreements (SLAs) and IT Service Management (ITSM), supporting your response with a relevant example. 7
2. a) Explain the main components of the Linux file system. Describe how file permissions and security are managed in Linux, and use a suitable example to support your explanation. 8
- b) Describe the key features of Windows Server operating systems. Examine the roles and functions of Active Directory and Domain Controllers within a Windows Server environment. 7
3. a) Define the term *protocol*. Discuss the purpose of the TCP/IP model and compare it with the OSI protocol stack, highlighting their key differences. 7
- b) Define information security. Describe its core principles, and examine best practices for securing servers and networks, supporting your answer with a relevant example. 8
4. a) Describe the different types of network network models. Discuss the core concepts and functions of a Domain Name System (DNS) server, and use a suitable example to support your explanation. 7
- b) Define virtualization and explain its key benefits. Provide a suitable example to illustrate its use. In addition, describe the different types of hypervisors, supporting your explanation with relevant examples. 8

5. a) Define cloud computing. Describe and distinguish between the various cloud service models, supporting your answer with suitable examples. 7
- b) Describe the key characteristics of scripting languages. Explain how these languages are used to automate common tasks, and provide relevant examples to support your answer. 8
6. a) Explain the role of configuration management tools and discuss how DevSecOps principles and practices enhance the secure software development lifecycle. 8
- b) Discuss how organizations can design and implement a structured troubleshooting approach to enhance system administration practices. 7
7. Write short notes on: (**Any two**) 2×5
 - a) Types of IT services
 - b) Linux File Ownership
 - c) Server Virtualization